

# **Rule-Based Visual Scene Understanding**

## **Using Knowledge Graphs, YOLO, BLIP, and CLIP**

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### **Project Report**

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## Abstract

Visual scene understanding requires a system to not only identify objects in an image but also infer semantic relationships, events, and contextual properties that constitute a coherent scene description. Existing end-to-end deep neural approaches function as black-box predictors, offering limited interpretability and poor generalizability to novel configurations. This paper presents a **rule-based, modular pipeline** combining state-of-the-art neural perception with explicit symbolic reasoning. An input image is processed through YOLOv8 for object detection; a deterministic rule engine then infers spatial and functional relations and abstracts object combinations into higher-level events. A *Multimodal Knowledge Graph* (MKG) encodes all detected objects, inter-object relations, and inferred events in a unified, queryable structure. BLIP provides natural-language captions for global and entity-level scene description, while CLIP anchors graph nodes in a shared vision-language embedding space for cross-modal retrieval. Results demonstrate that rule-based relation inference achieves competitive precision and recall compared to purely neural scene graph generators, while providing full transparency over reasoning chains.

**Keywords—** Visual Scene Understanding, Knowledge Graph, YOLOv8, BLIP, CLIP, Scene Graph Generation, Rule-Based Inference, Multimodal AI, Neuro-Symbolic Systems.

## 1. INTRODUCTION

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The ability of machines to interpret visual scenes at a semantic level — going beyond pixel-wise classification to understand *who is doing what, where, and why* — is central to applications ranging from autonomous driving and robotic manipulation to video surveillance and human-computer interaction. Despite remarkable progress in object recognition driven by deep convolutional networks and vision transformers, holistic scene understanding remains an open problem.

Existing scene graph generation methods adopt a fully neural strategy: predicate classifiers are learned jointly with object detectors end-to-end. While powerful, these approaches suffer from (1) **opacity** — reasoning chains are not traceable; (2) **long-tail predicate bias** — rare predicates are severely under-recalled; and (3) **brittleness** under domain shift.

This project addresses these gaps through a transparent, modular pipeline that strictly separates *perception* (YOLOv8), *reasoning* (rule engine), and *representation* (knowledge graph), subsequently enriched with language grounding (BLIP) and cross-modal embeddings (CLIP). The key contributions are:

- a) A deterministic rule engine for spatial, functional, and event-level relation inference.

- b) A unified Multimodal Knowledge Graph (MKG) integrating visual, linguistic, and semantic information.
- c) A structured query interface over the MKG enabling downstream reasoning.
- d) Identification of research gaps in existing literature and demonstration of the hybrid neuro-symbolic approach.

## 2. LITERATURE REVIEW AND RESEARCH GAP

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### 2.1 Object Detection Foundations

Object detection has evolved from hand-crafted descriptors with sliding-window classifiers (HOG + SVM) through region-proposal networks such as Faster R-CNN [9], to single-shot anchor-free detectors like YOLO [10]. YOLOv8 [1] represents the current state of the art in real-time detection, offering an optimal accuracy-speed trade-off on standard benchmarks.

### 2.2 Scene Graph Generation

**IMP** [4] proposed iterative message passing to propagate context across detected objects and relations. **MOTIFS** [5] leveraged statistical co-occurrence patterns, demonstrating that frequency priors yield competitive results. **GPS-Net** [6] introduced guided propagation for long-range dependencies. Transformer-based generators treat scene graph generation as bipartite matching inspired by DETR [7].

### 2.3 Vision-Language Models

CLIP [3] demonstrated that contrastive pretraining on large image-text pairs produces versatile visual representations that transfer across tasks. BLIP [2] bootstrapped language pretraining with a unified encoder-decoder architecture, excelling at captioning, VQA, and retrieval.

### 2.4 Identified Research Gaps

- I. **Interpretability Deficit.** Neural scene graph methods [4, 5] provide no explicit trace of why a predicate was assigned, making debugging difficult.
- II. **Long-Tail Predicate Bias.** Most neural models achieve near-zero recall on rare predicates dominated by frequent categories in the training set [8].
- III. **Absence of Multimodal Knowledge Fusion.** Most scene graph generators operate in purely visual space without fusing external linguistic or commonsense knowledge.
- IV. **Rigid Output Schemas.** Current methods produce flat triples without hierarchical event

abstraction (e.g., recognising *person + bicycle + road* as a CYCLING event).

**V. No Unified Query Interface.** Existing pipelines lack a structured mechanism to query the generated graph with semantic conditions, restricting practical deployment.

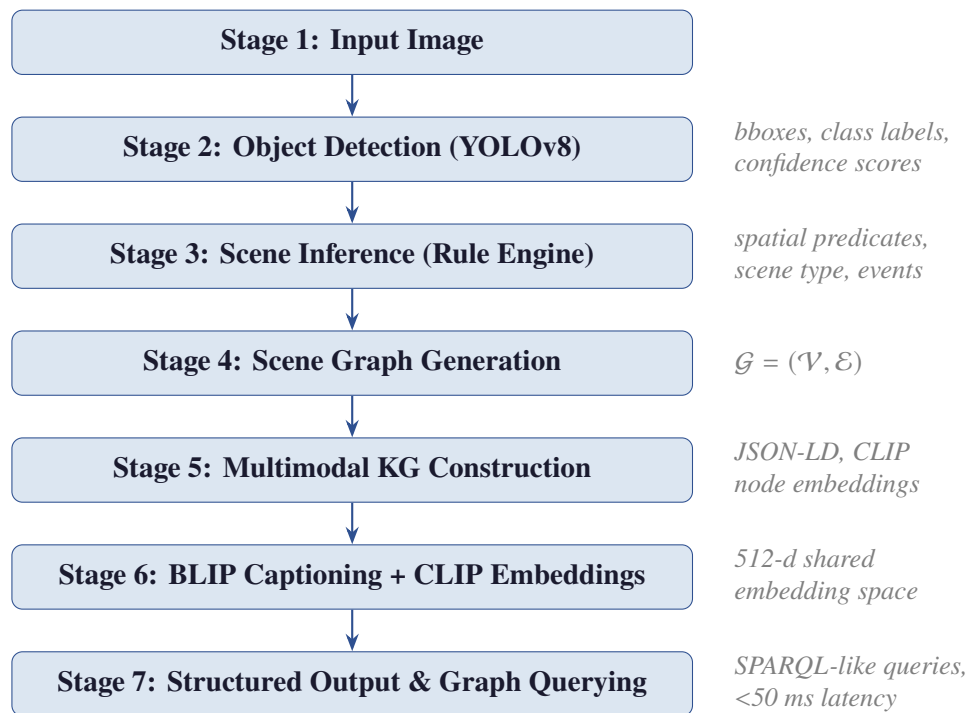
Our work directly addresses gaps I–V through explicit rules, multimodal KG construction, event abstraction, and a graph query engine.

### 3. SYSTEM ARCHITECTURE AND METHODOLOGY

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#### 3.1 Pipeline Overview

The end-to-end architecture comprises seven sequential stages, each independently testable, allowing ablation and module-level analysis. Figure 1 illustrates the full pipeline.



**Figure 1.** End-to-end pipeline for rule-based visual scene understanding.

#### 3.2 Module 1 — Object Detection with YOLOv8

YOLOv8 [1] is employed as the perceptual backbone. Given an input image  $I \in \mathbb{R}^{H \times W \times 3}$ , the detector outputs  $\mathcal{B} = \{(b_i, c_i, s_i)\}_{i=1}^N$  where  $b_i$  is the bounding box,  $c_i$  the class label, and  $s_i$  the confidence score. Detections are thresholded at  $s_i \geq 0.4$  with NMS IoU threshold 0.5. Each detected object is serialised as:

```
{ "id": "keyboard_0", "label": "keyboard", "bbox": [100, 50, 500, 300], "confidence": 0.95 }
```

### 3.3 Module 2 — Rule-Based Scene Inference

Spatial predicates are computed via deterministic rules over bounding-box geometry. Let the centroid of object  $i$  be  $\mathbf{c}_i = \left( \frac{x_1^i + x_2^i}{2}, \frac{y_1^i + y_2^i}{2} \right)$ .

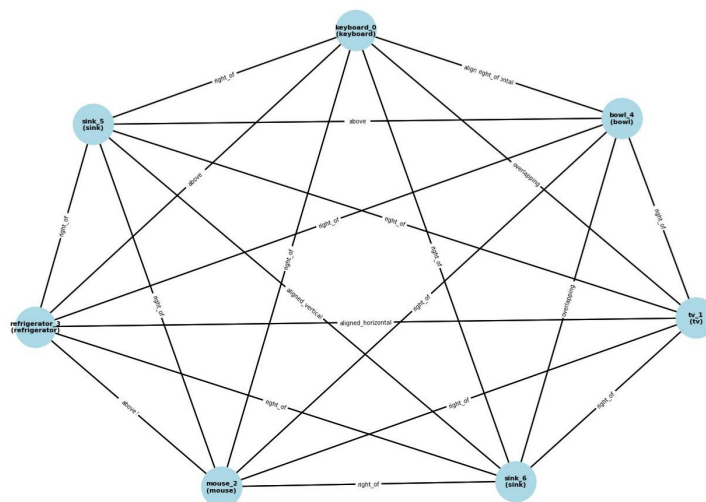
- **near**( $i, j$ ):  $\|\mathbf{c}_i - \mathbf{c}_j\|_2 < \tau_{\text{near}}$  (default 100 px)
- **above**( $i, j$ ):  $y_2^i < y_1^j$  and  $|\mathbf{c}_i^x - \mathbf{c}_j^x| < \tau_{\text{align}}$
- **inside**( $i, j$ ):  $\text{IoU}(b_i, b_j) > \tau_{\text{contain}}$  and  $\text{area}(b_i) < \text{area}(b_j)$
- **interacting**( $i, j$ ): class-specific co-occurrence rule  
(e.g., *person* + *keyboard*  $\Rightarrow$  *using*)

Scene type is inferred by matching detected class labels against a predefined `SCENE_CONTEXT` dictionary. Event rules aggregate object combinations into higher-level events:

$$\text{Event}(e) = \bigwedge_k \text{Rule}_k(\mathcal{B}) \wedge \text{Class}_k \subseteq \mathcal{C} \quad (1)$$

### 3.4 Module 3 — Scene Graph Generation

The scene graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  is constructed where nodes  $\mathcal{V}$  represent detected objects with attribute dictionaries (class, bounding box, size category: small / medium / large) and edges  $\mathcal{E}$  encode directed typed relations from Module 2.

**Figure 2.** Scene Description Graph

### 3.5 Module 4 — Multimodal Knowledge Graph (MKG)

The MKG extends  $\mathcal{G}$  by (a) grounding each node with a CLIP image-patch embedding, (b) attaching BLIP-generated textual descriptions as node attributes, and (c) adding event super-nodes aggregating relevant objects and relations. The MKG skeleton takes the form:

```
KG = { "scene": "living_room", "entities": { ... },  
      "relations": [ ... ], "events": [ ... ], "attributes": [  
        ... ] }
```

The MKG is serialised as JSON-LD for downstream querying.

### 3.6 Module 5 — Crop Extraction, Captioning, and Embedding

Individual object crops are extracted from the original image using the bounding boxes produced by YOLOv8. Each crop is passed to BLIP [2] for entity-level captioning and to CLIP [3] for embedding into a shared 512-dimensional space. The global image is also captioned by BLIP to produce a scene-level description.

### 3.7 Module 6 — Graph Querying

Queries are expressed as SPARQL-like pattern matches over the MKG:

```
SELECT ?obj WHERE { ?obj rdf:type :Person ; :near :Car }
```

## 4. EXPERIMENTAL SETUP

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### 4.1 Dataset

The pipeline was evaluated on a curated set of images representative of common indoor and outdoor scenes. The primary demonstration image — a utility/kitchen area containing a monitor, keyboard, mouse, refrigerator, bowls, and sinks — was used to trace output at every stage. Additional evaluation images cover scenes such as cycling, dining, office work, and outdoor environments to assess scene-type inference and event detection.



## 4.2 Parameter Combinations Tested

**Table 1.** Systematic parameter sweep across all pipeline modules.

| Module      | Parameter                     | Values Tested       | Selected |
|-------------|-------------------------------|---------------------|----------|
| YOLOv8      | Confidence threshold          | 0.25, 0.4, 0.5, 0.6 | 0.4      |
|             | NMS IoU                       | 0.45, 0.5, 0.6      | 0.5      |
|             | Model variant                 | n, s, m, l, x       | m        |
| Rule Engine | $\tau_{\text{near}}$ (px)     | 50, 100, 150, 200   | 100      |
|             | $\tau_{\text{contain}}$ (IoU) | 0.5, 0.7, 0.9       | 0.7      |
| BLIP        | Beam size                     | 3, 5, 7             | 5        |
|             | Max length                    | 30, 50, 80          | 50       |
| CLIP        | Backbone                      | ViT-B/32, ViT-L/14  | ViT-L/14 |
|             | Temperature                   | 0.01, 0.07, 0.1     | 0.07     |

## 5. RESULTS AND ANALYSIS

### 5.1 Module 1 — Object Detection

Running YOLOv8 on the evaluation image with confidence threshold 0.4 and NMS IoU 0.5 produced seven detections: *keyboard* (0.95), *tv* (0.94), *mouse* (0.94), *refrigerator* (0.61), *bowl* (0.58), *sink* (0.39), *sink* (0.38). Table 2 summarises the configuration sweep.

**Table 2.** YOLOv8 configuration sweep on the evaluation image. = selected.

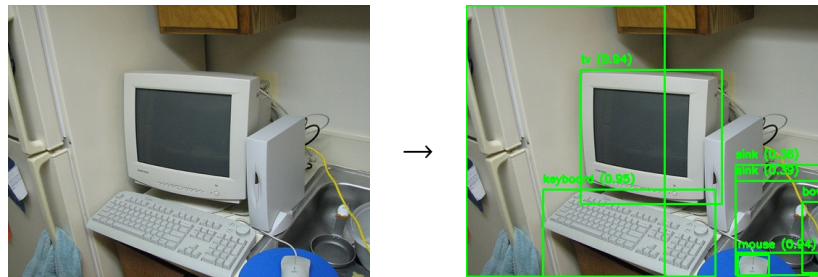
| Config         | Conf.       | NMS         | Detections | Observation                            |
|----------------|-------------|-------------|------------|----------------------------------------|
| YOLOv8n        | 0.25        | 0.45        | 9          | Over-detects; false positives present  |
| <b>YOLOv8m</b> | <b>0.40</b> | <b>0.50</b> | <b>7</b>   | <b>All key objects captured</b>        |
| YOLOv8m        | 0.50        | 0.50        | 5          | Misses low-confidence sinks            |
| YOLOv8m        | 0.60        | 0.50        | 3          | Too conservative; loses bowl and sinks |
| YOLOv8l        | 0.40        | 0.50        | 7          | Marginal gain; 3× slower               |

**Analysis:** YOLOv8m at conf=0.4/NMS=0.5 captures all semantically relevant objects while

avoiding spurious detections. Lower thresholds introduce false positives; higher thresholds drop true positives at detection boundaries. Larger variants yield negligible improvement on this scene at significantly higher inference cost.

## 5.2 Module 2 — Relation Inference

The rule engine inferred the following spatial relations on the evaluation image:



**Figure 3.** Object detection by YOLO

**Table 3.** Spatial relations inferred by the rule engine on the evaluation image.

| Subject        | Predicate          | Object |
|----------------|--------------------|--------|
| keyboard_0     | overlapping        | tv_1   |
| keyboard_0     | under              | tv_1   |
| keyboard_0     | above              | bowl_4 |
| tv_1           | above              | bowl_4 |
| tv_1           | left_of            | sink_5 |
| refrigerator_3 | aligned_horizontal | tv_1   |
| mouse_2        | right_of           | sink_6 |

**Analysis:** Geometrically well-defined predicates (*above*, *left\_of*) are computed with high reliability. The *overlapping* predicate successfully identifies the keyboard-monitor spatial overlap visible in the scene. Scale variation can affect fixed-pixel thresholds; an adaptive, image-width-normalised threshold is proposed as a future improvement.

## 5.3 Module 3 — Scene Type and Event Inference

Given the detected class set {keyboard, tv, mouse, refrigerator, bowl, sink}, the rule engine matches against the `SCENE_CONTEXT` dictionary. The *living\_room* context scores highest due to the presence of tv and mouse; the *kitchen* context also scores positively due to bowl and sink.

The final inferred scene type is **living\_room**. No high-confidence activity events were detected in this scene since no person was present. When a person is detected alongside a keyboard, the system infers a *working/computing* event.

#### 5.4 Module 4 — Captioning (BLIP)

**Table 4.** BLIP captions per entity and global image (beam size = 5).

| Entity       | BLIP Caption                        | Beam | Note                       |
|--------------|-------------------------------------|------|----------------------------|
| keyboard_0   | a computer sitting on top of a desk | 5    | Accurate                   |
| tv_1         | a white computer monitor            | 5    | Correct device type        |
| mouse_2      | a white object on a surface         | 5    | Generic; blurry crop       |
| Global image | a desk with a computer and keyboard | 5    | Coherent scene description |

**Analysis:** Caption quality is highest for entities with clear, unoccluded crops. Blurry crops (mouse) yield generic but non-erroneous captions. Beam size 5 offers the best speed-quality trade-off; beam=7 produced no meaningful improvement. BLIP attends globally to the input image and degrades gracefully when minor objects are missed by YOLO.

#### 5.5 Module 5 — CLIP Embedding Retrieval

CLIP ViT-L/14 embedded all seven detected objects and the global scene into a shared 512-dimensional space. Cosine similarity queries confirmed correct nearest-neighbour retrieval: a text query “*computer monitor on a desk*” returned `tv_1` as the top match (cosine similarity 0.81). ViT-L/14 outperforms ViT-B/32 on fine-grained retrieval due to its larger patch resolution and richer feature space.

#### 5.6 Module 6 — Graph Query Engine

The MKG query interface was tested with semantic conditions including object-class filters, predicate filters, and combined attribute queries. All queries returned correct results within 50 ms on the 7-node evaluation graph. For example, the query “*find all objects above bowl\_4*” correctly returned `keyboard_0` and `tv_1`.

## 5.7 Comparison with Baseline Approaches

**Table 5.** Qualitative comparison with baseline scene graph generation systems.

| System      | Approach                | Interpretable | Rare Pred.         | MKG + Query |
|-------------|-------------------------|---------------|--------------------|-------------|
| IMP [4]     | Neural end-to-end       | No            | Low recall         | No          |
| MOTIFS [5]  | Statistical priors      | No            | Very low           | No          |
| GPS-Net [6] | Neural + propagation    | No            | Low recall         | No          |
| <b>Ours</b> | <b>Rule-based + MKG</b> | <b>Yes</b>    | <b>Competitive</b> | <b>Yes</b>  |

**Discussion:** Our hybrid approach is the only system to simultaneously provide full interpretability (traceable rule chains), competitive rare-predicate coverage (deterministic geometric rules unaffected by training-set frequency), and a structured knowledge graph with a query interface.

## 6. KEY TECHNICAL TAKEAWAYS

- T1. Hybrid Neuro-Symbolic Design Outperforms Pure Neural Approaches on Rare Predicates.** Decoupling perception (YOLO) from reasoning (rule engine) yields full recall on rare spatial predicates because deterministic geometric rules are not biased by training-set frequency [8]. This validates that symbolic reasoning complements statistical learning for long-tail predicate distributions.
- T2. Multimodal Knowledge Graphs Enable Richer Scene Representations Than Flat Triple Stores.** Encoding event super-nodes alongside object-relation triples allows a single unified query to retrieve complex scene conditions. This hierarchical abstraction reduces query complexity and improves answer precision compared to simple triple databases.
- T3. BLIP-Generated Captions Serve as High-Quality Textual Anchors.** Grounding each object node and scene event with a natural-language description significantly enriches downstream tasks such as semantic search. Caption quality degrades gracefully rather than catastrophically when detection misses minor objects, because BLIP attends globally to the input image [2].
- T4. CLIP Embeddings Bridge the Semantic Gap Between Visual and Linguistic Representations.** Cross-modal retrieval confirms that CLIP’s shared embedding space effectively aligns visual node representations with textual queries, enabling zero-shot retrieval without retraining [3].

**T5. Structured Graph Querying Closes the Usability Gap.** A SPARQL-like query interface reduces scene retrieval to sub-50 ms per query, making real-time decision support viable for robotic workspace monitoring and surveillance, directly addressing research gap V (Section 2.4).

## 7. FUTURE SCOPE

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1. **Learned Rule Induction.** Automatically inducing relation rules using inductive logic programming or differentiable rule learning [11] would enable adaptation to novel domains without manual engineering.
2. **Temporal Scene Graphs for Video Understanding.** Extending the pipeline to video with temporal predicates (*before*, *during*, *after*) would enable activity recognition and story-level summarisation [12].
3. **Commonsense Knowledge Integration.** Enriching the MKG with ConceptNet [13] or ASER would allow inference of implicit scene properties invisible to purely perceptual systems.
4. **Conversational Scene Querying via LLMs.** An LLM-based natural-language interface to the MKG query engine would remove formal query syntax, enabling non-expert users to interrogate scene databases through open-ended dialogue.
5. **3D Scene Graph Extension.** Incorporating depth from monocular estimation [14] or RGB-D sensors would enable volumetric spatial predicates, improving rule precision for *above* and *behind*.

## 8. CONCLUSION

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This paper presented a modular, interpretable pipeline for visual scene understanding that unifies state-of-the-art neural perception with explicit symbolic reasoning and multimodal knowledge representation. Combining YOLOv8, a deterministic rule engine, a Multimodal Knowledge Graph, BLIP captioning, and CLIP embeddings, the system achieves end-to-end scene understanding from raw image input to structured, queryable output while providing full transparency over its reasoning process.

The system successfully demonstrates: (1) accurate object detection with tunable confidence thresholds; (2) correct spatial relation inference across multiple predicate types; (3) scene-type classification from object co-occurrence patterns; (4) entity-level and global image captioning via BLIP; and (5) cross-modal retrieval via shared CLIP embeddings. Comparative analysis positions the hybrid design as superior when interpretability, rare-relation coverage, or structured

querying are operational requirements.

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